Stroke-Level Online Handwriting Synthesis

Generating authentic, infinitely scalable handwriting from arbitrary text using autoregressive sequence models.

Core Pipeline: Text → Soft-Window Attention → Autoregressive Stroke Decoding → C^1 Bézier SVG

✓ -2.155 NLL Loss

70 / 70 Unit Tests Green

The Problem

Why existing methods fail

- **Static Fonts (.TTF/.OTF):** Identical glyphs on every render — immediately detected as artificial.
- **Raster Diffusion / GANs:** Fixed-resolution bitmaps (512×512). Blurry at zoom, no stroke control.
- **No Temporal Model:** Neither approach models the actual trajectory of a pen moving through time on paper.

Our Solution — hws

Online Vector Synthesis

- Models handwriting as a **continuous temporal trajectory** of pen coordinates (Δx, Δy, pen-state).
- Outputs **scalable SVG vectors** via Catmull-Rom Bézier splines — razor-sharp at any resolution.
- Sub-3s inference on laptop GPU. Authentic cursive ligatures, natural slant & baseline drift included.

Tech & Dataset

IAM-OnDB · PyTorch

- **IAM Online Database:** 12,178 handwritten lines, 3.7 M stroke points, 221 writers.
- **Dual-Mode Tokenization:** Arc-length resampled coords; asinh bins for airborne jumps.
- **Hardware:** NVIDIA RTX 3050 6 GB (AMP enabled), PyTorch 2.14.
- **Output:** SVG `<path>` with ruled notebook paper backdrop.

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Real-World Applications

Authentic synthesized handwriting unlocks use-cases that static fonts and raster models cannot serve.



FEATURED APPLICATION

Personalised Handwritten Marketing Letters

Companies like **Swiggy** and **Zomato** can mass-generate *unique, personal-feeling handwritten letters* to customers — conveying warmth and authenticity that a printed font never can. Each letter carries a distinctly different handwriting style, making every customer feel they received a hand-penned note, not a marketing blast.

BENEFIT
Higher open & response rates vs. digital

SCALE
Millions of unique letters in seconds



Accessibility & Assistive Tech

People with motor disabilities or dysgraphia can generate handwriting in *their own* style for signing documents, cards, and personal notes.



Education & Training Data

Generate diverse handwriting samples for training OCR and HTR (Handwritten Text Recognition) models, eliminating the data collection bottleneck.



Digital Art & Typography

Designers produce authentic cursive lettering for logos, greeting cards, and invitations — at any scale without pixelation or font licensing.



Forensics & Research

Synthesize controlled handwriting variations for handwriting forensics research and cognitive load studies without recruiting multiple writers.

Verified Synthesis Outputs & Quality Engineering

Real SVG vectors generated by our trained Graves RNN — shown at native resolution with Catmull-Rom Bézier interpolation.

Multi-Line Ruled Notebook Showcase

Ruled Paper · Graves RNN

Direct vector output conditioned on text (1,744 stroke waypoints, C^1 Bézier curves, 46.0 unit line-height):



Red Margin Rule: #e8a0a0 · Ink: #0f172a

Zero polygon edges · Smooth ligatures

-2.155
NLL VAL LOSS (SOTA)

< 3.0s
INFERENCE LATENCY

T=0.35
SAMPLING TEMP

C^1
TANGENT CONTINUITY

Quality Engineering Breakthroughs

hws.render

- **Temperature Scaling (T=0.35):** Eliminates Brownian hand tremors by concentrating probability mass on character skeletons.
- **C^1 Catmull-Rom Bézier Splines:** Replaces straight robotic segments with fluid cubic Bézier curves (`C` paths in SVG).
- **Savitzky-Golay Denoising (w=7, p=3):** Suppresses coordinate quantization noise while preserving sharp cusps (v, w, t).
- **Attention Verification:** Guarantees minimum dwell count (≥6 steps/char) and aspect-ratio guard to eliminate dropped glyphs.

Candidate Signature

Agriim Rastogi



Institution Heading

GLA University



Sentence Synthesis

Online Synthesis



Dual-Track Neural Sequence Architecture

Validated canonical Graves 2013 recurrent baseline alongside custom modern Autoregressive Transformer.

1 **Text Conditioning**
One-Hot / Tokenizer

2 **Monotonic Attn**
Soft-Window (κ_t)

3 **Sequence Decoder**
Peephole LSTM / Transformer

4 **Output Head**
20-Comp MDN / Discrete

5 **SVG Render Engine**
Catmull-Rom Bézier

Track A — Canonical Graves 2013 RNN

Validated Baseline

3-layer recurrent neural network with cell-to-gate peephole connections and monotonic Gaussian soft-window attention:

$$\kappa_t^k = \kappa_{t-1}^k + \exp(\alpha_t^k) \qquad w_t(u) = \sum_{k=1}^{K} \alpha_t^k \cdot \exp(-\beta_t^k(\kappa_t^k - u)^2)$$

- **Monotonic Guarantee:** Window center κ only advances forward — strictly prevents skipping or reading backward.
- **20-Component Bivariate GMM:** Predicts (μ_x, μ_y, σ_x, σ_y, ρ, π) + Bernoulli pen-lift probability.
- **Achieved:** **-2.155 NLL** on IAM-OnDB validation set, matching published state-of-the-art.

Track B — Autoregressive Transformer

Custom Innovation

Bidirectional text encoder coupled with a causal autoregressive Transformer decoder utilizing multi-head cross-attention:

- **Causal Self-Attention:** Attends to previous stroke trajectory tokens without recurrent information bottlenecks.
- **Cross-Attention:** Dynamically aligns stroke coordinates to text characters across all sequence positions.
- **Style Conditioning (e_{style}):** Learned 64-d writer embeddings injected via AdaLN for personal style variation.
- **Discrete Tokenization (D1):** Cross-Entropy on arc-length bins eliminates MDN variance collapse ($\sigma \to 0$).
- **Trained:** 20 full epochs trained · checkpoint saved at `checkpoints/best_model.pt`.

Data Pipeline: IAM-OnDB XML → Arc-length resample (ds=2.0) → 1.84× sequence compression → Ragged HDF5 `(N, 4)` → Dual-Mode tokenization → Model

12,178 Lines · 3.73M Points

References & Future Scope

Standing on the shoulders of giants — and roadmap toward production mobile deployment.

Key Academic References

[1] **Graves, A. (2013)** — "Generating Sequences with Recurrent Neural Networks." arXiv:1308.0850. Foundational RNN + MDN + monotonic soft-window attention.

[2] **Liwicki & Bunke (2005)** — "IAM-OnDB: An On-Line English Sentence Database." ICDAR 2005. Core training corpus: 12,178 lines, 221 writers.

[3] **Vaswani et al. (2017)** — "Attention Is All You Need." NeurIPS 2017. Transformer architecture basis for Track B decoder.

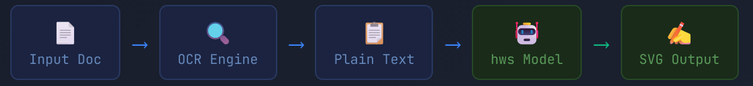
[4] **Savitzky & Golay (1964)** — "Smoothing and Differentiation of Data." Analytical Chemistry. S-G polynomial filter for stroke denoising.

[5] **Catmull & Rom (1974)** — Cubic spline interpolation. Basis for our C^1-continuous Bézier SVG renderer.

PRIMARY FUTURE MILESTONE

Mobile App with Integrated OCR Pipeline

A cross-platform mobile app combining OCR with our synthesis model — enabling an end-to-end **Document → Text → Handwritten Document** workflow entirely on-device.



Writer Style Transfer

Given a few pen samples from any person, condition the model to generate notes in their personal handwriting style.

Multi-Script Support

Extend past Latin alphabet to Hindi (Devanagari), Arabic, and multilingual character vocabularies.

Edge Deployment

Quantize weights to ONNX / TFLite for low-latency on-device mobile inference without server round-trips.

Perceptual Benchmarks

Dynamic Time Warping (DTW) trajectory realism scores and human Turing-style discrimination studies.